The documentation and process conversion measures necessary to comply with this revision shall be completed by 10 Dec 93.

INCH-POUND

MIL-S-19500/253D
10 September 1993
SUPERSEDING
MIL-S-19500/253C
27 September 1989

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER
TYPE 2N930, JANTX

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, low-power transistors. One level of product assurance is provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

P_T		V_{CBO}	V_{EBO}	V_{CEO}	I_C	T_J and T_{STG}
T_A = +25°C 1/	T_C = +25°C 2/					
mW	mW	V dc	V dc	V dc	mA dc	°C
300	600	60	6	45	30	-65 to +200

1/ Derate linearly at 2.0 mW/°C above T_A = +25°C.
2/ Derate linearly at 4.0 mW/°C above T_C = +25°C.

1.4 Primary electrical characteristics.

	h_{FE1}	h_{FE2}	C_{obo}	$\|h_{fe}\|$	$V_{BE(sat)}$ 1/	$V_{CE(sat)}$ 1/
	V_{CE} = 5 V dc I_C = 10 µA dc	V_{CE} = 5 V dc I_C = 500 µA dc	I_E = 0 V_{CB} = 5 V dc 100 kHz ≤ f ≤ 1 MHz	I_C = 500 µA dc V_{CE} = 5 V dc f = 30 MHz	I_C = 10 mA dc I_B = 0.5 mA dc	I_C = 10 mA dc I_B = 0.5 mA dc
			pF		V dc	V dc
Min	100	150		1.5	0.6	
Max	300		8.0	6.0	1.0	1.0

1/ Pulsed; see 4.5.1.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Research Laboratory Electronic and Power Source Directorate, ATTN: AMSRL-EP-RD, Fort Monmouth, NJ 07703-5601 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, and figure 9, T-2A.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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4.3 Screening. Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX level
11	I_{CBO1} and h_{FE1}.
12	See 4.3.1.
13	Subgroup 2 of Group A; ΔI_{CBO1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±25% of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

$V_{CB} \geq 20$ V dc; P_T = 300 mW at T_A = +30°C ±5°C.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX) of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps and footnotes of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVb (JANTX) of MIL-S-19500.

Subgroup	Method	Condition
B3	1027	V_{CB} = 20 V dc; T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. P_T = 300 mW, No heat sink or forced-air cooling on the devices shall be permitted.
B4	2037	Test condition A: all internal leads for each device shall be pulled separately.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps and footnotes of table I, group A, subgroup 2 herein.

4.4.3.1. Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
C3	2036	Test condition E.
C6	1026	V_{CB} = 20 V dc; T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. P_T = 300 mW, No heat sink or forced-air cooling on the devices shall be permitted.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector to base breakdown voltage	3001	Bias condition D; I_C = 10 µA dc	$V_{(BR)CBO}$	60		V dc
Emitter to base breakdown voltage	3026	Bias condition D; I_E = 10 µA dc	$V_{(BR)EBO}$	6.0		V dc
Collector to emitter breakdown voltage	3011	Bias condition D; I_C = 10 mA dc pulsed (see 4.5.1)	$V_{(BR)CEO}$	45		V dc
Collector to emitter cutoff current	3041	Bias condition B; V_{CE} = 45 V dc	I_{CES1}		2.0	nA dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 5 V dc	I_{CEO}		2.0	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO}		5.0	nA dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 45 V dc	I_{CBO}		10	nA dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 µA dc	h_{FE1}	100	300	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 500 µA dc	h_{FE2}	150		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 mA dc; Pulsed (see 4.5.1)	h_{FE3}		600	
Saturation voltage and resistnace	3071	I_C = 10 mA dc; I_B = 0.5 mA dc; Pulsed (see 4.5.1)	$V_{CE(sat)}$		1.0	V dc
Base emitter voltage (saturated)	3066	Test condition A; I_C = 10 mA dc; I_B = 0.5 mA dc; Pulsed (see 4.5.1)	$V_{BE(sat)}$	0.6	1.0	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High - temperature operation		T_A = +175°C				
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 45 V dc	I_{CES2}		10	µA dc
Low - temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 µA dc; Pulsed (see 4.5.1)	h_{FE4}	20		
Subgroup 4						
Small-signal short-circuit forward- current transfer ratio	3206	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 1 kHz	h_{fe}	150	600	
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 5 V dc; I_C = 500 µA dc; f = 30 MHz	$\|h_{fe}\|$	1.5	6.0	
Open circuit output capacitance	3236	V_{CB} = 5 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8.0	pF
Noise figure	3246	V_{CE} = 5 V dc; I_C = 10 µA dc; R_g = 10 kΩ;	NF			
Test 1		f = 100 Hz			5	dB
Test 2		f = 1 kHz			3	dB
Test 3		f = 10 kHz			3	dB
Subgroups 5 and 6						
Not applicable						

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 7						
Small-signal open circuit output admittance	3216	V_{CB} = 5 V dc; I_E = 1.0 mA dc; f = 1 kHz	h_{ob}	0.0	1.0	µmho
Small-signal open circuit reverse-voltage transfer ratio	3211	V_{CB} = 5 V dc; I_E = 1.0 mA dc; f = 1 kHz	h_{rb}	0.0	6×10^{-4}	
Small-signal short-circuit input impedance	3201	V_{CB} = 5 V dc; I_E = 1.0 mA dc; f = 1 kHz	h_{ib}	25	32	Ω

1/ For sampling plan see MIL-S-19500.

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4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Lead formation and finish as specified (see 3.3.1).

b. Type designation and product assurance level.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, AV, EA, MI, SM
Navy - AS, CG, MC
Air Force - 13, 85, 99
NASA - NA
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/253D	2. DOCUMENT DATE (YYMMDD)
3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER, TYPE 2N930, JANTX		

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

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a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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